$^{15}N_2$ uptake in *Brachypodium distachyon* treated with biodegradable hydrogels in EcoFAB model systems

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Introduction

Biological nitrogen fixation (BNF) is a process by which microbes process biologically-available nitrogen (N) from atmospheric nitrogen (N_2). N_2 is an inert gas and unavailable for direct plant uptake. In BNF, either symbiotic or free living microbes convert atmospheric N_2 to NH_3. This is further nitrified into NO_3- and NH_4+, which can be taken up by plant roots and arbuscular mycorrhizal fungi (AMF). About 50% of the global N budget can be attributed to BNF (Chalk 2016), rendering this an important process for N cycling and and agricultural productivity. Stable isotopes like $^{15}N_2$ can be used to trace N movement through soil, microbial, and plant systems.

N-fixing bacteria can be classified as either symbiotic (residing in root nodules) or free living in the rhizosphere. Common free living N-fixers include *Azotobacter vinelandii* and *Azospirillium brasilense*, which have been shown to increase plant root growth and nitrogen uptake (Santi et al, 2013). AMF are plant root mutualists and enhance N uptake to the plant, possessing genes of primary N assimilation and catabolism (nitrate reductase, glutamine synthetase and glutamate dehydrogenase, Pfeffer et al, 2005). They also contribute to bacterial-fungi interactions with rhizospheric bacteria, as they offer microniches for bacteria to inhabit and act as 'superhighways' for them. Bacteria alter soil composition and engage in signaling which affect AMF communities (Deveau et al, 2018)

EcoFABs (fabricated ecosystems) are bench scale models designed to observe these types of rhizophere interactions. They have been previously used for microbial compatibility testing in BioBead (Mafune et al, 2024) and are well-suited for studying plant root, microbial, mycorrhizal fungal interactions. This study will address $^{15}N_2$ movement from the atmosphere to soil microorganisms to a plant host. A previous pilot trial (Davis, 2025) demonstrated that wheat plants (*Triticum aestivum*) grow poorly in the EcoFAB chambers. This follow up study will use the model plant *Brachypodium distachyon* with BioBeads treated with microbial consortia with strong N fixation capabilities. In this study, the air around *Brachypodium distachyon* plants will be enriched with $^{15}N_2$ for short period of time. The tissues will be analyzed for ^{15}N content relative to ^{14}N ($\delta^{15}N$). This information will provide insight on the N fixing capabilities of BioBead hydrogels containing microbial consortia. The null hypothesis is that there is no difference in mean $\delta^{15}N$ content between the BioBead and control plants. The alternative hypothesis is that there is a difference in mean $\delta^{15}N$ content between the BioBead and control plants.

Materials and Methods

Experimental Design

To evaluate the effects of microbial inoculation on plant growth and nitrogen fixation, we prepared two types of hydrogel beads: BioBeads, containing a mix of arbuscular mycorrhizal fungi (AMF), *Azospirillum brasilense*, and *Azotobacter vinelandii*, and control beads, which lacked microbial inoculation. A total of 50 mL of each bead type was produced for the experiment according to methods in Mafune et al (2024).

Six EcoFAB 2.0 units were placed into 3 plant growth chambers. Each chamber lid was modified by drilling a single hole using a drill press and a 7/18" drill bit, into which a 20 mm rubber butyl stopper was inserted. Since each chamber could accommodate 2 EcoFAB 2.0 units, each BioBead unit was paired with a control to mitigate environmental effects.

Seed Preparation and Germination
Brachypodium distachyon seeds were surface sterilized by submerging them in 70% ethanol for 1 minute, followed by a 5-minute immersion in chloramine T. The seeds were then rinsed 10 times with MilliQ water to remove any residual sterilant. To promote uniform germination, the sterilized seeds were placed on a 1.7% agar plate and cold-stratified at 4°C for 48 hours before being incubated at 20°C until germination was observed.

EcoFAB Assembly and Inoculation
Prior to inoculation, the EcoFABs were assembled and bleach-sterilized to minimize contamination. Under sterile conditions, each EcoFAB was filled with 7 mL of 0.5× Murashige & Skoog (MS) medium without nitrate. Approximately 40–50 beads (either BioBeads or control beads) were added to the bottom of the sampling port, which was then covered with sterilized tinfoil. The EcoFABs were incubated at 20°C with a 12-hour light/12-hour dark photoperiod inside clean tote bins. Liquid media was refreshed weekly by adding 2 mL of 0.5× MS medium.

$^{15}N_2$ Contaminant Screening

To screen for potential ammonium contaminants (2014 ref), 30mL of the $^{15}N_2$ gas was acid-washed with 10mL of 1N hydrochloric acid (Doty et al., 2016) in a Balch tube, agitated for 180 seconds, neutralized to pH 6 and screened in the Gallery analyzer (ThermoFischer). This test was performed in triplicate.

$$NH_4^+ + HCl = NH_4Cl + H+$$

The concentration of N_2 is 0.134M:

$$\frac{0.03L}{224L/mol} = \frac{0.00134molN_2/30mL}{0.010LHCl} = 0.134M$$

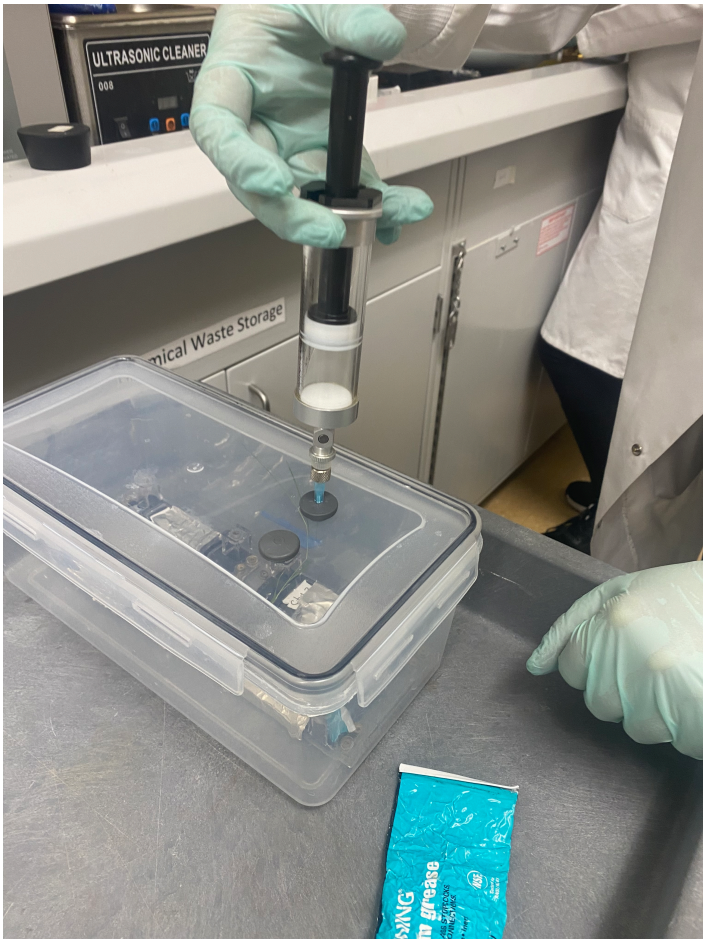
The instrument did not detect any NH_4+ (LOD 0.1 uM). This means that an NH_4+ contaminant, if present, would be below 0.0000746% or 1.8 ppb.

Sterile tinfoil was used to cover sampling ports as well as the root viewing pane to prevent any potential light damage to the study organisms.

Nitrogen Enrichment and Sample Processing
At six weeks post-inoculation, N enrichment was introduced by adding 60 mL of $^{15}N_2$ gas (98% purity, Cambridge Isotopes) to each container to achieve a 2% enrichment level (see Dosing Rationale below). The container lids were sealed with vacuum grease for a one-week exposure period.

The following week, the dosing was repeated but with a gradient:

- BioBead 1 and Ctrl 1: 60 mL
- BioBead 2 and Ctrl 2: 30 mL
- BioBead 3 and Ctrl 3: 0 mL



The plants were grown for 8 more weeks until harvest and processing.

Isotopic Mass Spectroscopy

At the end of the experiment, the entire plant (roots and above-ground biomass) were harvested and dried at 60°C for 72 hours. Elemental analyses and isotope composition were obtained from dried and powdered samples by a Sercon Integra2 by the University of Oklahoma Stable Isotope Lab.

Data Collection

Isotopic compositions are reported in ^{15}N values referenced to the stable N isotope composition of N_2 in air. The ^{15}N values are calculated using the following equation,

$\delta^{15}N = (R_{sa}/R_{std} - 1) * 1,000(‰)$,

where R_{std} and R_{sa} are the $^{15}N/^{14}N$ ratios of the standard and sample (Filley et al., 2001).

$^{15}N_2$ dosing rationale

For $^{15}N_2$ enrichment dosing, other research groups have used simple calculations of chamber volume or headspace percentage. Examples include 20% of atmosphere (Alexander 1989), 10% vol of chamber (Martinez-Feria et al., 2024), or 1% spike of headspace or chamber (Doty et al., 2016) without replacement. One study (Smercina 2019) used a different approach with pN2 1.87 atm. Wewalwela (2020) used 5% $^{15}N_2$ enrichment. We chose 2% as an enrichment target for the EcoFAB experiments.



Figure 1: Fig 2: EcoFAB units after inoculation, showing roots and hydrogel beads in liquid media.

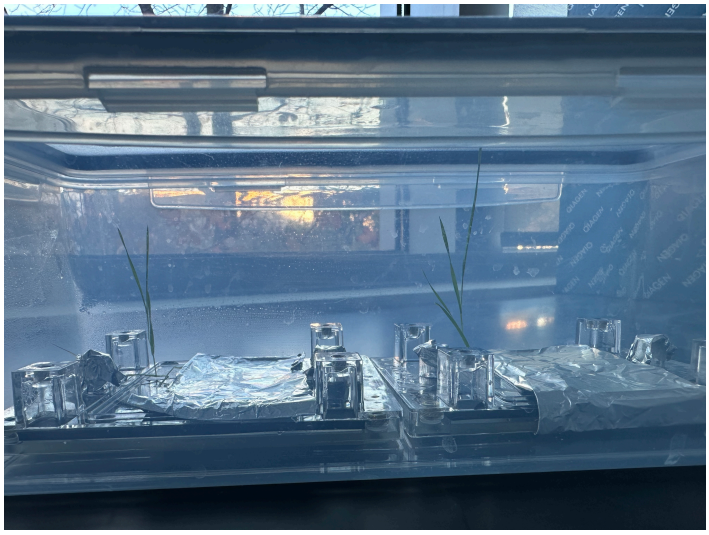


Figure 2: Fig 3: Sideview of EcoFAB units at week 3 showing plant growth.

The ideal gas law frames the relationships between pressure, volume, temperature and moles of gas.

$$PV = nRT$$

$$n = \frac{PV}{RT}$$

The ideal gas law allows calculations of moles of gas for the given 2.945L container housing EcoFABs.

$$n = \frac{PV}{RT} = \frac{1atm \times 2.945L}{0.0821 \frac{L \times atm}{mol \times K} \times 298K} = 0.121mol$$

Another calculation corrects the number of moles for 2% enrichment at 98% purity.

$$0.02 \times 0.121mol^{1}5N2 = \frac{0.0061mol^{1}5N2}{0.98} = 0.0062mol^{1}5N2$$

Using the ideal gas law, we can now use the new mole parameter to find the volume (and account for the mole fraction of nitrogen gas in air).

$$V = \frac{nRT}{P} = \frac{0.0062mol \times 0.0821 \frac{L \times atm}{mol \times K} \times 298K}{1atm} \times 0.78 = 0.120L$$

Observations

Slow early growth and nitrate supplementation

The seedlings showed slow early growth, including yellowing of leaves. To maintain plant health, 100 μL of NH_4NO_3 (825 mg/L) was added to each EcoFAB at week 2.

Confocal microscopy

A Yokogawa CSU-X1 spinning disk confocal microscope was used to image the hydrogel beads, roots, and media at week 10.

The micrographs from the BioBead units are teeming with life. Microbes can be seen in the media. Hyphae penetrate beads and roots.

Results

Ammonium concentrations in media over time

```
library(ggplot2)

# data entry
BioBead1 <- c(143.54,31.65,156.36,39.97,42.24,107.87,65.40,58.40,48.91,42.27,136.03)
BioBead2 <- c(44.64,32.61,0,0,0,0,0,0,0,0,21.77)
BioBead3 <- c(21.04,17.12,27.11,28.62,32.43,76.64,46.13,70.80,36.8,21.57,516.56) #note: last data point

Ctrl1 <- c(308.51,176.63,28.71,104.38,12.81,15.58,8.92,12.34,9.68,13.72,1.35)
Ctrl2 <- c(318.96,347.62,303.56,42.55,39.44,36.98,15.33,7.1,5.6,9.45,15.26)
Ctrl3 <- c(149.86,155.22,159.93,50.81,106.48,41.01,13.59,13.48,7.02,0,10.34)
```

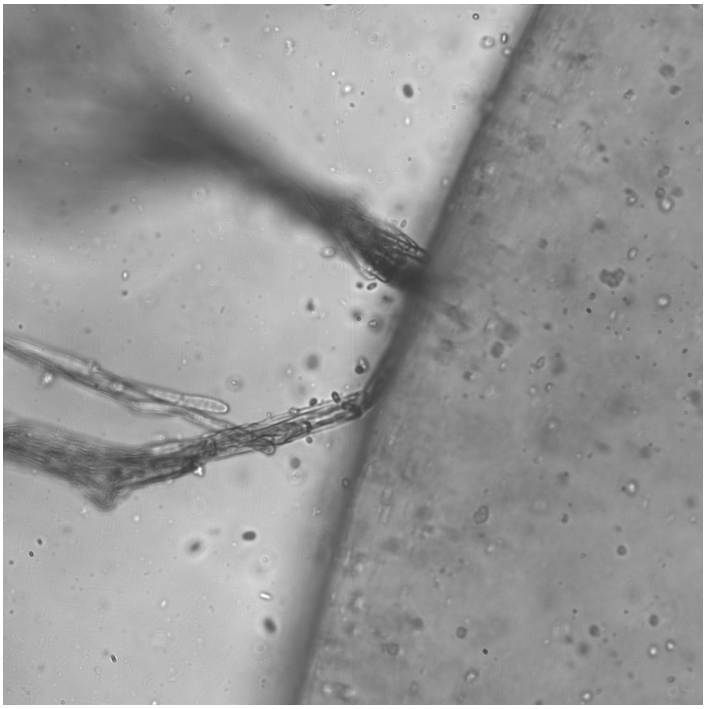


Figure 3: Fig 4: BioBead at 40x showing hyphae interacting with the bead, week 10.

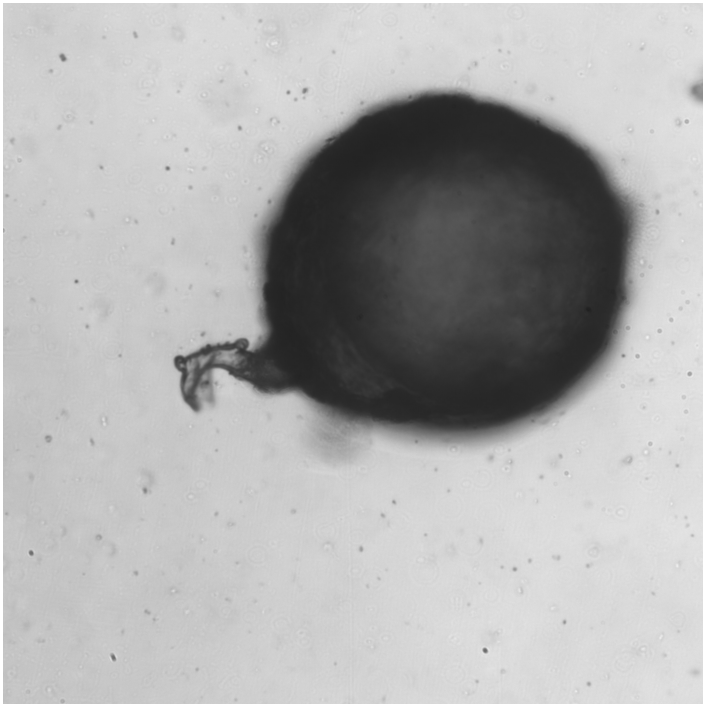


Figure 4: Fig 5: Germinating AMF spore from BioBead unit, 40x, week 10.

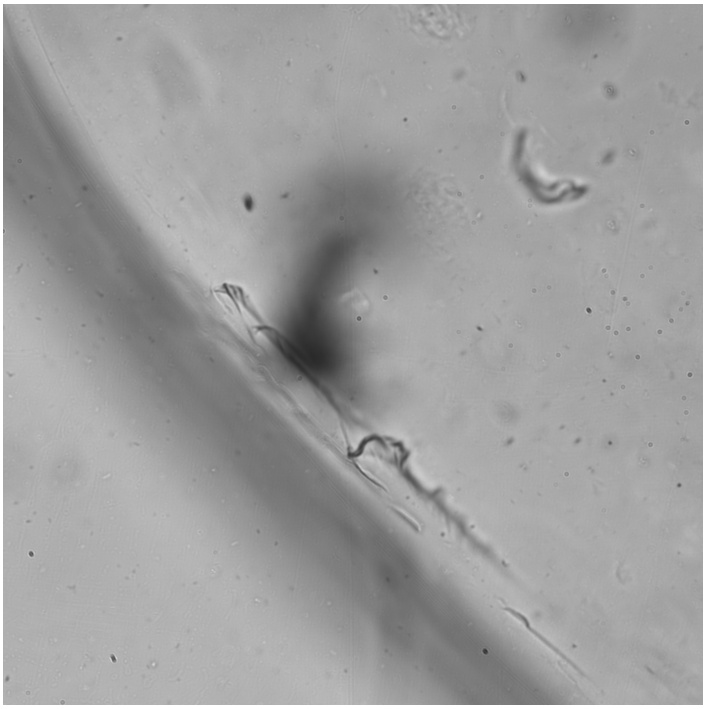


Figure 5: Fig 6: Control hydrogel bead at 40x. Week 10.

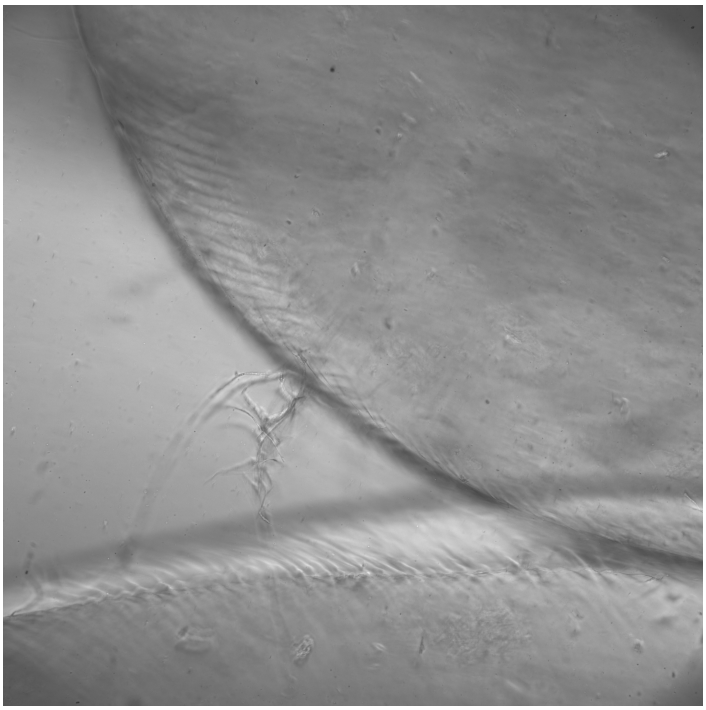


Figure 6: Fig 7: Control hydrogel bead from another unit, 40x. Week 10.

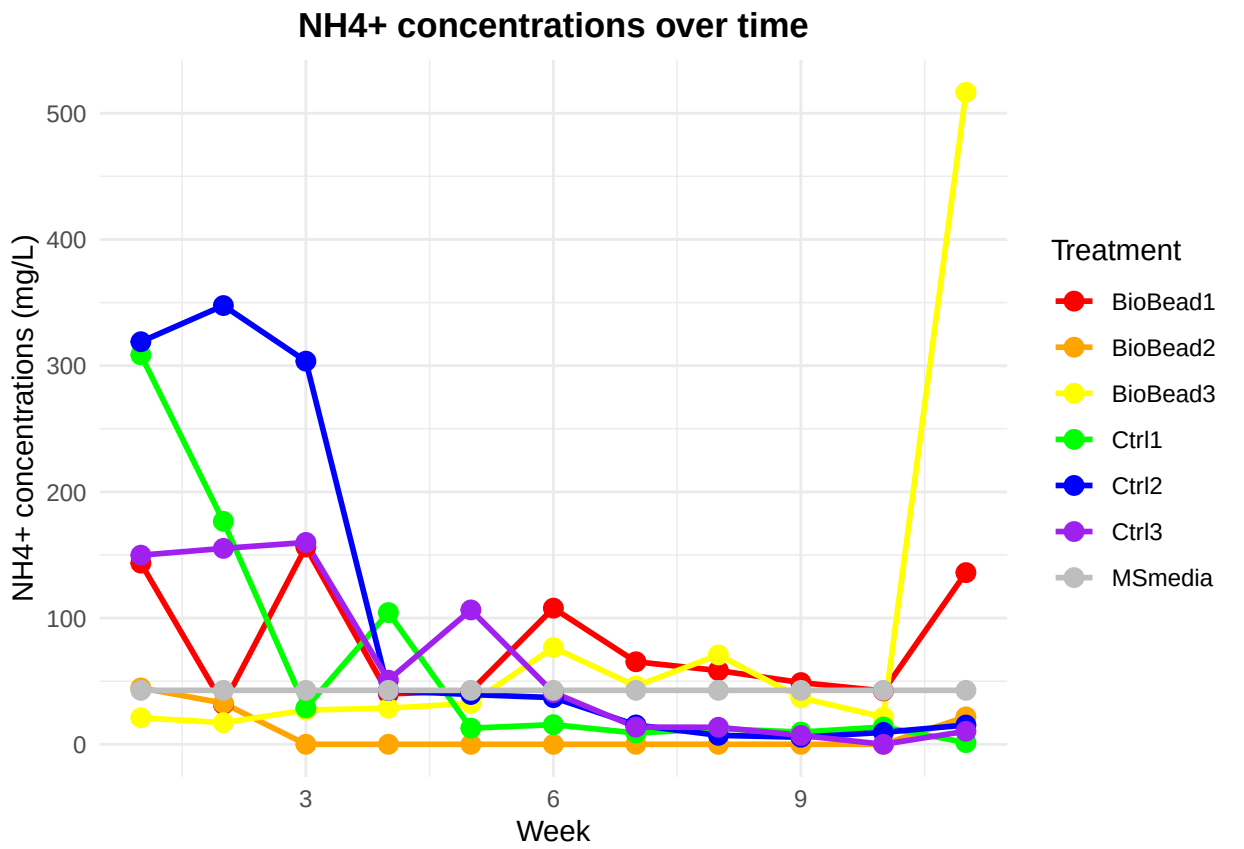
```r
MSmedia <- c(42.75,42.75,42.75,42.75,42.75,42.75,42.75,42.75,42.75,42.75,42.75)

# Creating the data frame with explicit week assignments
data <- data.frame(
  Week = rep(1:11, 7),  # Repeating weeks for each treatment
  Treatment = rep(c("BioBead1", "BioBead2", "BioBead3", "Ctrl1", "Ctrl2", "Ctrl3", "MSmedia"), each = 1
  NH4_Concentration = c(BioBead1, BioBead2, BioBead3, Ctrl1, Ctrl2, Ctrl3, MSmedia)
)

# Define custom colors
treatment_colors <- c("BioBead1" = "red", "BioBead2" = "orange", "BioBead3" = "yellow",
                      "Ctrl1" = "green", "Ctrl2" = "blue", "Ctrl3" = "purple",
                      "MSmedia" = "gray")

# Creating the line plot
ggplot(data, aes(x = Week, y = NH4_Concentration, group = Treatment, color = Treatment)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  scale_color_manual(values = treatment_colors) +
  labs(title = "NH4+ concentrations over time",
       x = "Week",
       y = "NH4+ concentrations (mg/L)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



There is also total organic nitrogen data that may be analyzed at a later date as well. The values are all negative, and do not fluctuate very much.

Bead and root imaging

At the end of experiment, the hydrogel beads and roots from each EcoFAB were stained according to methods published in Mafune, Kasson, and Winkler (2024): "B. distachyon plants were removed through the seed port and roots were set aside for clearing and staining techniques to observe any mycorrhizal establishment in fine roots. The roots were cleared and stained using modified methods from Phillips and Hayman (1970), which included a KOH concentration of 8% (w/v)…Then, roots were acidified in 2% HCl for 10 min, soaked for 6 h in 0.05% Trypan Blue in lactoglycerol, and finally destained in MilliQ H2O for 18–24 h to remove excess dye (Phillips and Hayman 1970). Samples from each EcoFAB were kept separate and observed under the Leica EZ4D dissecting scope for staining patterns." (5% KOH w/v was used in this experiment).

d15N concentrations

```
library(ggplot2)
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag

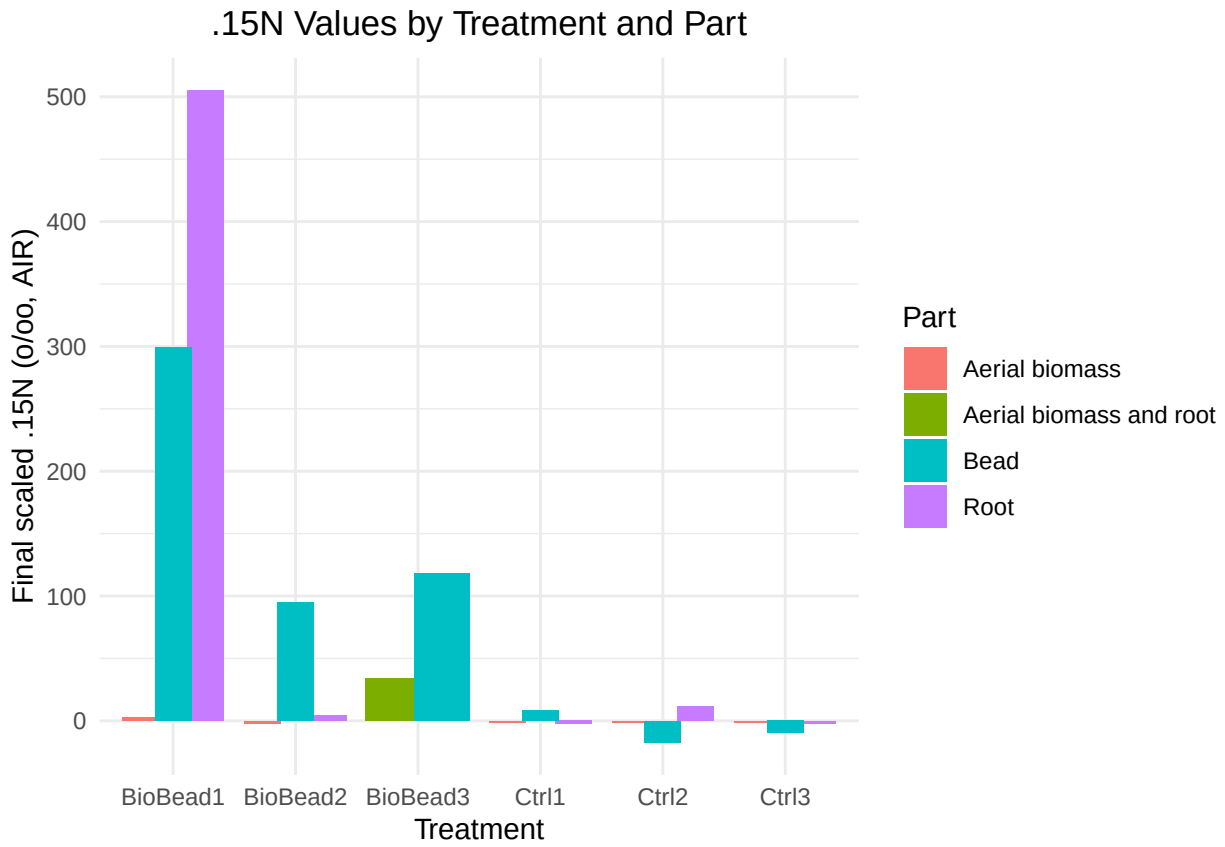
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# read in data
data <- read_csv("d15N_EcoFAB.csv")
```

```
## Rows: 17 Columns: 4

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): Sample ID, Name, Part
## dbl (1): d15N
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Create the plot
ggplot(data, aes(x = Name, y = d15N, fill = Part)) +
  geom_bar(stat = "identity", position = position_dodge((width = 0.8))) +
  labs(
    title = " 15N Values by Treatment and Part",
    x = "Treatment",
    y = "Final scaled  15N (%, AIR)",
    fill = "Part"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```



Overall, the BioBead samples yield greater $\delta^{15}N$ values compared to untreated controls. The bead and root portions contained more ^{15}N compared to aerial biomass. This suggests slow uptake of N into aerial biomass, which could have occurred for several reasons. Following ^{15}N enrichment, the plants were grown for 8 more weeks to allow for optimal N metabolism and uptake. However, plant growth was observed to be slow. In a hydroponic environment with limited gas exchange, tissue growth may not be optimal hence stifling N uptake. Alternatively, the plants may have needed longer than 8 weeks for uptake.

Due to low sample yield, the aerial biomass and roots from BioBead3 were combined. The analytical team also noted that Ctrl1 Bead, Ctrl2 Bead, Ctrl3 Root, Ctrl3 Bead, and BioBead2 Root all had beam areas outside the analytical range of the instrument or were beyond the within-run calibration. Therefore, concentrations and/or isotope values for these samples should be carefully evaluated.

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